Ex1: Operating systems

Answers:

1. Docker compose up is much faster than booting up a VM.
   a container can be ready in seconds while VM can take minute or more.
   The reason for the difference in boot times is:

   a. VM uses a hypervisor to emulate virtual hardware (CPU,GPU,RAM,etc ...), VM also need to be running through a virtual BIOS and initializing these virtual hardware components, when Docker containers run directly on the host hardware.

   b. A VM must load and boot an entire OS kernel from scratch and all the background system services. Docker containers share the hosts OS kernel. Starting a container is just starting a new isolated process on the host, skipping the OS boot entirely.

   c. VMs pre allocate a fixed chunk of CPU and RAM upon startup, which requires system overhead to reserve. Containers dynamically utilize the host system resources as needed, without heavy pre allocation on startup.

2. To send a new containerized app to a friend, there are some prerequisites:
   a. Docker installed.
   b. Docker-compose installed (if the app uses multiple services).
   c. Access to registry (docker.io, github, Artifactory, etc…).

   We have two ways for sending the app:
   a. You push your image to a registry using the command docker push.
   And the receiver does docker pull and then docker run to the image.
   b. You send them the folder containing the Dockerfile and the source code,
   And the receiver needs to docker build and run on their machine.

3. When you run a Docker stack containing an app and a proxy, there is no separate OS kernel for each container.
   Both the app container and the proxy container share the same OS kernel.
   Docker uses Linux kernel features like Namespaces (to isolate what a process can see, like PID and networks) and cgroups (to limit what a process can use, like CPU and memory) to keep the app and the proxy isolated from one another.

The differs from running two separate VMs:

if you ran the app and the proxy on two separate VMs, the architecture would be entirely different:

a. The host computer would run a hypervisor.
b. VM1(app) would boot and run its own complete, heavy OS kernel.
   VM2(Nginx) would boot and run its own separate OS kernel(Duplicated OS kernels).